African Wildllife Image Classification Pipeline

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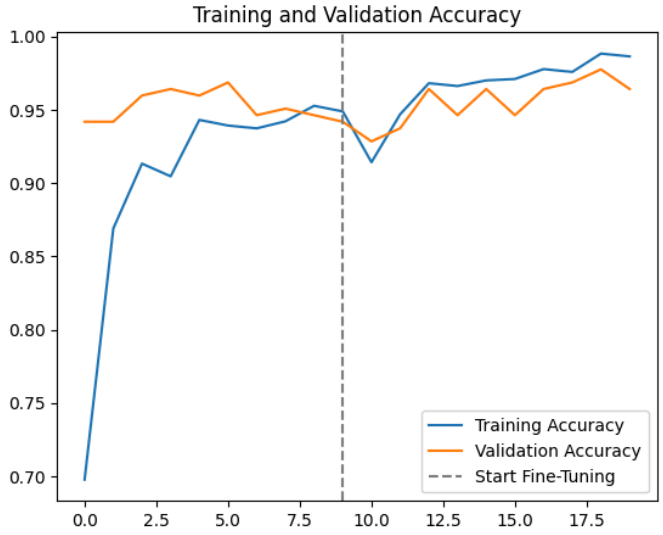
1) Objectives and Methodology

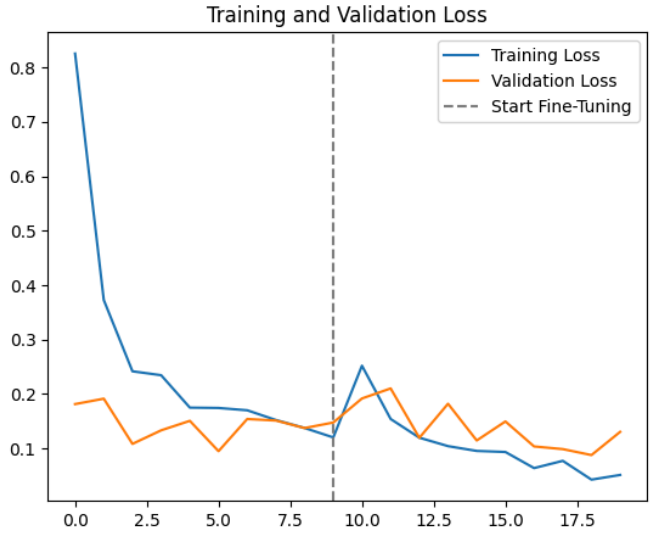
This project implements a two-stage transfer learning pipeline to classify the Big Five wildlife species using a pre-trained **ResNet50 convolutional neural network.** The primary objective is to evaluate the practical impacts of feature extraction versus fine-tuning. Data ingestion was handled via tf.data pipelines, partitioned into a 70/15/15 split for training, validation, and testing configurations. The input images were standardized to 224x224 spatial dimensions. To mitigate overfitting caused by limited data, a localized augmentation pipeline was applied exclusively to the training set.

2) Structural Changes & Network Architecture

The base ResNet50 architecture was modified to accommodate the specific dataset by replacing its original fully connected layers with a custom classification head. The modified architecture includes a global average pooling layer to flatten the spatial dimensions, followed by a dense layer. Dropout layers were incorporated to reduce overfitting. The final classification layer utilizes a Dense layer with 5 nodes and a softmax activation function to output prediction probabilities for the five wildlife species.

3) Performance Evaluation (Feature Extraction vs Fine-Tuning)



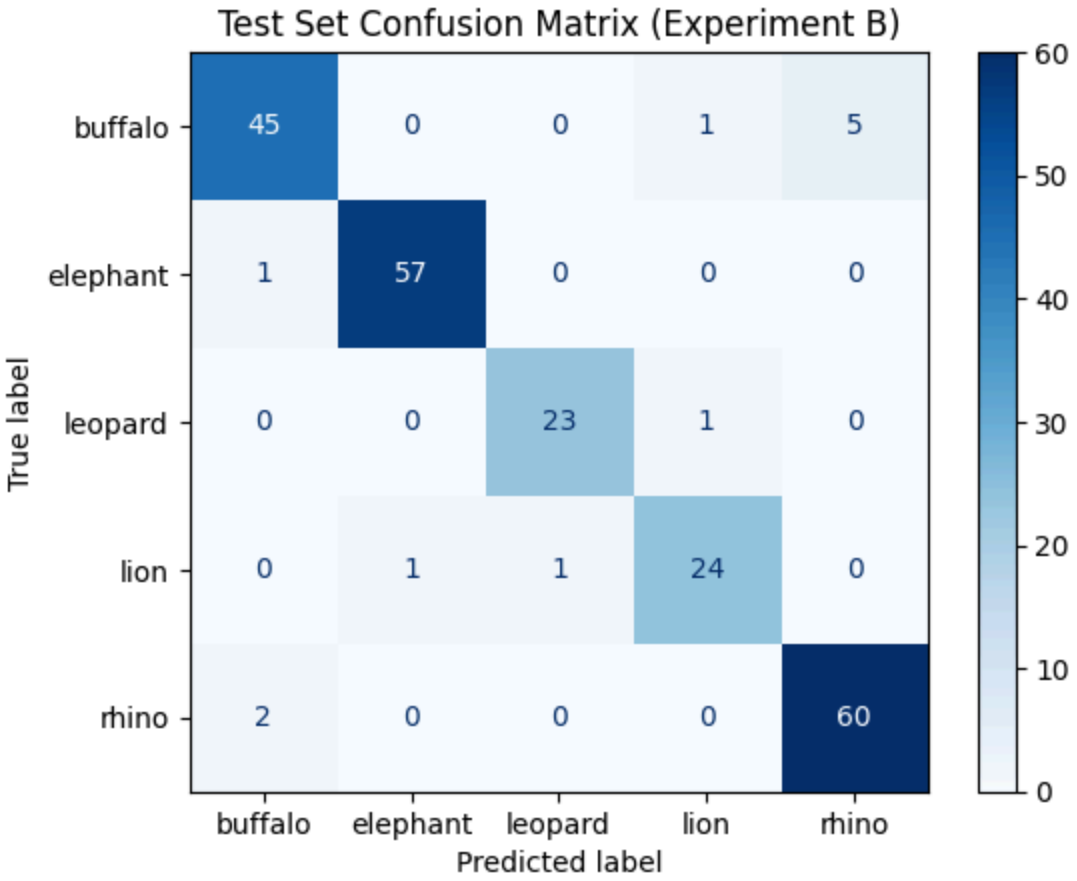


The model's performance was evaluated through two distinct training approaches on the validation and test sets:

- Experiment A (Feature Extraction): The convolutional base of the ResNet50 model was frozen, and only the custom classification head was trained. This phase ran for 10 epochs with a learning rate of 0.001 to leverage pre-trained features, establishing a baseline validation accuracy of 96.88% and a test set F1-score of 0.95.
- Experiment B (Fine-Tuning): The top layers of the convolutional base were unfrozen and trained alongside the classification head to allow the network to learn complex, dataset-specific features. Fine-tuning improved the peak validation accuracy to 97.77%, demonstrating that adapting the network's shape detectors provides a measurable performance gain.

The transition to fine-tuning caused a momentary spike in validation loss as the weights adjusted, which was followed by a rapid recovery. Diverging training and validation loss curves observed in the final epochs (17–19) indicated the onset of model overfitting, confirming that 10 epochs of fine-tuning served as the optimal stopping point for this dataset.

4) Classification Metrics and Error Analysis



The final fine-tuned model achieved an overall accuracy of 95% on the test set, demonstrating strong generalization capabilities.

- Class Reliability: According to the F1-Score, Elephants proved to be the most reliable class (0.97), while Buffaloes were the least reliable (0.93). The model achieved a 0.98 recall rate for Elephants, successfully identifying 98% of the actual elephant photographs.
- Morphological Confusion: The highest misclassification rate occurred between the heavily built herbivores. The model incorrectly predicted 5 actual buffaloes as rhinos, and 2 actual rhinos as buffaloes.
- Anomalous Errors: The model generated low-frequency errors, such as misidentifying 1 actual buffalo as a lion and 1 actual lion as an elephant.
- Macro Average Validation: The final unweighted macro average score across all five classes was 0.95 (95%). This metric prevents the larger sample sizes of buffalo and rhino images from artificially inflating the overall evaluation score.

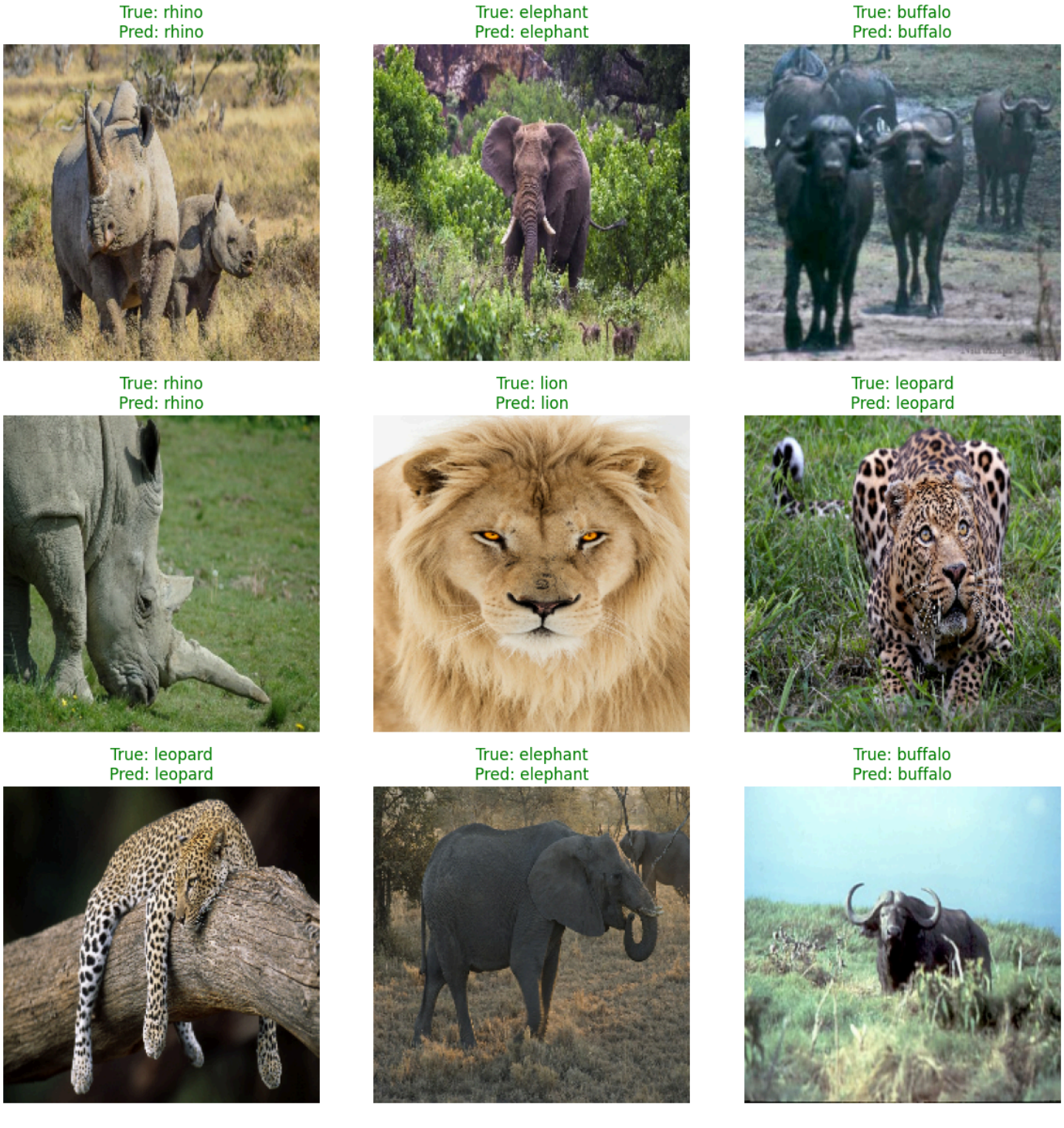
5) Result Matrix

Metric Evaluated	Experiment A (Feature Extraction)	Experiment B (Fine-Tuning)
Best Validation Accuracy	96.88%	97.77%

Metric Evaluated	Experiment A (Feature Extraction)	Experiment B (Fine-Tuning)
Test Set Macro F1-Score	0.95	0.94
Average Epoch Training Time	27.37 sec	26.56 sec

6. Test Set Visual Predictions

To validate the model's feature detection capabilities, a random batch of unseen test images was passed through the fine-tuned pipeline (Experiment B). The visual matrix below confirms the model's classification accuracy under various morphological, occlusion, and lighting conditions.



Project Repository & Source Code

The complete, reproducible codebase—including data ingestion pipelines, model architecture, training loops, and evaluation metrics—is available in the associated Google Colab environment.

You can find the complete code for this project in the Google Colab notebook here.

You can also access the project repository on GitHub here.